PRE-CLASS PREP

Blood Transfusions & ABGs

Know Before You Go

Read this before class — it will make the lecture click faster.

Big picture: Two distinct topics covered today — blood transfusion safety and acid-base interpretation. Both require pattern recognition and systematic thinking. ABGs are read the same way every time; learning the steps now makes it automatic.

BLOOD PRODUCTS

- **Packed RBCs:** restore blood levels without substantially increasing volume
- **Whole blood:** used for significant blood loss (trauma, surgery)
- **Plasma:** liquid portion containing clotting factors
- **Platelets:** stop or prevent bleeding; used in transplants, cancer, surgery
- **Universal donor:** O− · **Universal recipient:** AB+
- Blood type and crossmatch required before transfusion

BEFORE & DURING TRANSFUSION

- Signed consent; pre-transfusion vital signs
- Two-nurse check before beginning
- IV size ideally 18–20 gauge; **infuse with normal saline only**
- RN stays with client for **first 15 minutes** — most common time for reaction
- Vital signs at 15 min after start, then minimum hourly
- Must complete within **4 hours** of leaving blood bank
- If reaction: **stop transfusion immediately**, keep IV open with NS, remain with client, send blood bag and tubing to blood bank

ABG NORMAL VALUES & INTERPRETATION STEPS

pH
7.35 – 7.45
<7.35 = acidosis
>7.45 = alkalosis

CO_2
35 – 45 mmHg
Respiratory component

HCO_3
22 – 26 mEq/L
Metabolic component

1. **Look at pH** — acidosis (<7.35) or alkalosis (>7.45)?
2. **Use ROME** — Respiratory Opposite, Metabolic Equal. If pH and CO_2 move in opposite directions → respiratory. If pH and HCO_3 move in same direction → metabolic.
3. **Identify compensation** — if all three values are abnormal → partially compensated. If pH is back in normal range → fully compensated.

THE FOUR ABG DISORDERS

Respiratory Acidosis — pH ↓ · CO_2 ↑
Lungs retaining too much CO_2
Causes (DEPRESS): Drugs, Edema, Pneumonia, Respiratory center damage, Emboli, Spasm (asthma), Sac elasticity (COPD)
S&S: shallow respirations, confusion, fatigue, headache, cyanosis
Tx: improve ventilation; O_2; bronchodilators; naloxone if opioid-related; may need intubation if CO_2 >50

Respiratory Alkalosis — pH ↑ · CO_2 ↓
Lungs losing too much CO_2
Causes (TACHypnea): Temperature elevation, Anxiety, Conditions causing ↓O_2, Hypoxemia/hyperventilation
S&S: rapid breathing, tingling in extremities, light-headedness, tachycardia
Tx: treat cause; keep calm; coach slow breathing; fall precautions

Metabolic Acidosis — pH ↓ · HCO_3 ↓
Kidneys have too little bicarb / too much H+
Causes (MUDPILES): Malnutrition, Uremia, DKA, Poisoning, Isoniazid, Lactic acidosis, Ethanol, Severe diarrhea
S&S: Kussmaul respirations, fruity breath (DKA), confusion, flushed skin
Tx: treat cause; IV bicarb if severe; seizure precautions; monitor K^+ closely

Metabolic Alkalosis — pH ↑ · HCO_3 ↑
Kidneys have too much bicarb / too little H+
Causes: excess antacids, diuretics, vomiting, prolonged NG suction, hyperaldosteronism, blood transfusions
S&S: hypoventilation, tingling, muscle twitching, confusion
Tx: treat cause; IV fluids; replace K^+; antiemetics; monitor for respiratory distress

COMPENSATION — HOW THE BODY RESPONDS

- **Lungs respond first** — fast (minutes to hours); control CO_2; if acidotic → blow off CO_2; if alkalotic → retain CO_2
- **Kidneys respond second** — slow (hours to days); control H+ and HCO_3; if acidotic → reabsorb HCO_3 and secrete H+; if alkalotic → reabsorb H+ and secrete HCO_3
- When one organ causes the problem, the other compensates — this is why all three values may be abnormal in partially or fully compensated states